

Solid Waste Pollution

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Learning Objectives

After taking this class, the students will be able to:

- *understand* what solid waste pollution is and its consequences
- *acknowledge* the waste management practices of dealing with the solid waste pollution
- *master* waste sorting in practice

2

What is solid waste pollution?

3

Solid Waste Pollution

- Solid waste is useless and sometimes hazardous material with low liquid content.
- Solid wastes include municipal garbage, industrial and commercial waste, sewage sludge, wastes resulting from agricultural and animal husbandry operations and other connected activities, demolition wastes and mining residues.



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Classification of Solid Waste Pollution

Types of Solid Waste Pollution	Sources
Industrial solid waste	Including mining slag, smelting slag, chemical waste slag, fuel slag, industrial waste, construction waste slag, and other industrial waste slag
Urban domestic waste	Food waste, paper, plastic, glass, metal, fabric, etc.
Electronic waste	Household appliances, information technology and telecommunications equipment, lighting devices, power tools, medical equipment, vending machines
Construction waste	Including soil, rocks, mortar, concrete debris, broken bricks, waste pipes, and materials generated from decoration and renovation, various packaging, and other wastes, etc.
Hazardous wastes	Flammable, explosive, toxic, and substances that are reactive, corrosive, and infectious
Municipal Sludge	River silt, human excreta, etc.
Agricultural solid waste	Straw, vegetable offcuts, and livestock and poultry manure

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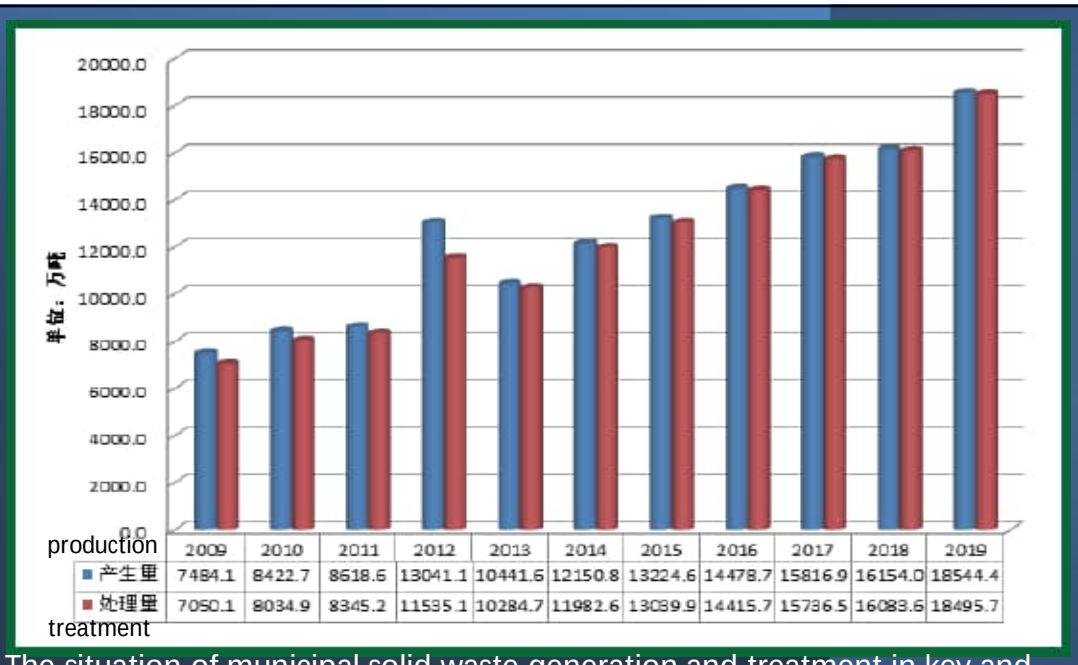
An Overview of Solid Waste Production

Year	General industrial solid wastes	Industrial hazardous wastes	Medical wastes	Urban domestic wastes	Covered cities
2020	138000	4498.9	84.3	23560.2	196
2019	155000	4643	81.7	21147.3	200
2018	131000	4010.1	78.1	20194.4	202
2017	148000	3344.6	72.1	18850.5	214
2016	191000	2801.8	68.9	18564.0	246
2015	192000	2436.7	62.2	16816.1	244
2014	238306.23	2937.05	54.75	16148.81	263

Unit:10000 Ton

(National Annual Report on Environmental Pollution Control of Solid Waste in Major Cities, 2014-2020)

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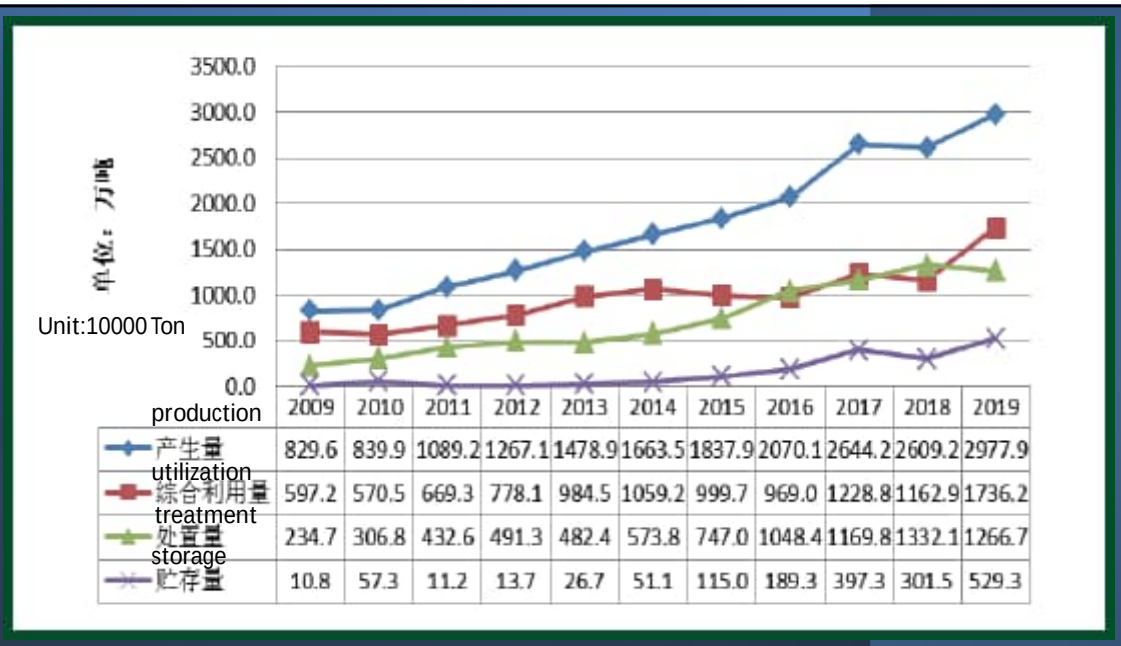
The situation of municipal solid waste generation and treatment in key and model cities (2009-2019)

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The generation and treatment of medical waste in key and model cities (2009-2019)

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production	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
产生量	829.6	839.9	1089.2	1267.1	1478.9	1663.5	1837.9	2070.1	2644.2	2609.2	2977.9
utilization 综合利用量	597.2	570.5	669.3	778.1	984.5	1059.2	999.7	969.0	1228.8	1162.9	1736.2
treatment 处置量	234.7	306.8	432.6	491.3	482.4	573.8	747.0	1048.4	1169.8	1332.1	1266.7
storage 贮存量	10.8	57.3	11.2	13.7	26.7	51.1	115.0	189.3	397.3	301.5	529.3

Annual report of generation, utilization, treatment, and storage of industrial hazardous waste in key and model cities (2009-2019)

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Secondary Pollution and Health Hazard

Solid waste and air pollution



Solid waste and water pollution



solid waste and soil pollution



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How do we manage solid waste?

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Reflective Question:

Can you name some practices regarding solid waste management?

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Waste Management Priority

Generation → Collection → Transport → Disposal

- Waste management is based on a certain order of priority.
  - Primarily, the generation of waste should be avoided.
  - If waste is generated, it must be prepared for reuse or reused.
  - If reusing is not possible, waste must be primarily recovered as materials (recycled), and secondarily recovered as energy.
  - Waste may be disposed of in landfills only if its recovery is not technically or financially feasible.



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Recycling in Sweden





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Recycling in Finland

- When you are going to live in Finland, waste recycling is one of the basic living skills that you need to learn. https://www.infofinland.fi/en/housing/waste-and-recycling





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Recycling in Finland

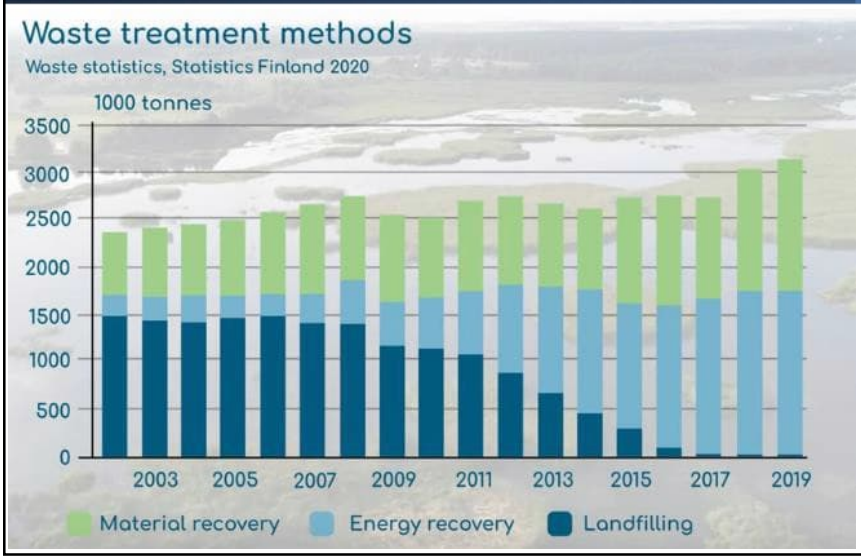






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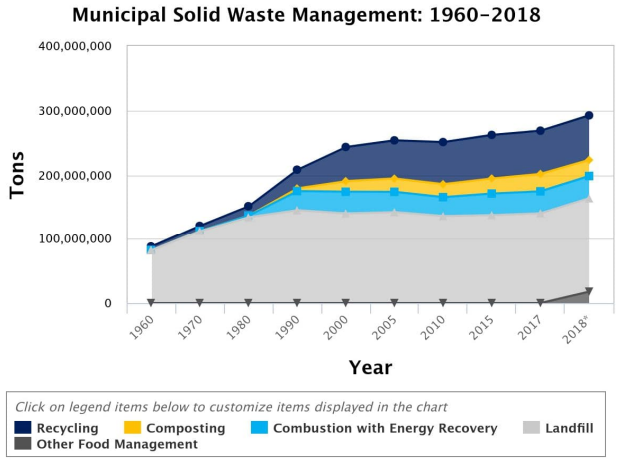
Recycling in Finland



SYKE Suomen ympäristökeskus, 2021 Waste Management in Finland

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Waste Management in the US





How San Francisco Became a Global Leader in Waste Management

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Solid Waste Management in China



Filled up, the largest domestic landfill is about to be closed



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Solid Waste Management in China

Statistics of China's standards, policies, and regulations on solid waste from 2003 to 2018.

Number of documents	2013	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
National standards	2	0	3	0	11	16	9	12	24	17	8	9	8	13	36	13
Industry standards	2	2	2	16	10	9	19	5	18	12	20	18	21	16	16	32
Local standards	3	1	0	3	1	2	5	4	10	13	15	36	20	21	32	15
Policies and regulations	48	88	89	112	151	374	646	589	658	499	345	364	338	305	428	422

(Guo et al, 2021)

中华人民共和国固体废物污染环境防治法

2020-04-30　　字号：[大] [中] [小]　[打印]

4月29日，十三届全国人大常委会第十七次会议审议通过了修订后的固体废物污染环境防治法，自2020年9月1日起施行。全文如下。

中华人民共和国固体废物污染环境防治法

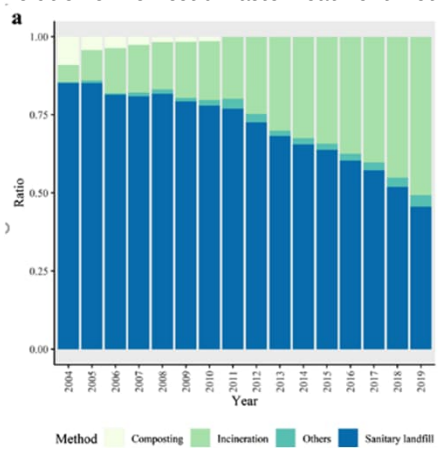
（1995年10月30日第八届全国人民代表大会常务委员会第十六次会议通过　2004年12月29日第十届全国人民代表大会常务委员会第十三次会议第一次修订　根据2013年6月29日第十二届全国人民代表大会常务委员会第三次会议《关于修改〈中华人民共和国文物保护法〉等十二部法律的决定》第一次修正　根据2015年4月24日第十二届全国人民代表大会常务委员会第十四次会议《关于修改〈中华人民共和国港口法〉等七部法律的决定》第二次修正　根据2016年11月7日第十二届全国人民代表大会常务委员会第二十四次会议《关于修改〈中华人民共和国对外贸易法〉等十二部法律的决定》第三次修正　2020年4月29日第十三届全国人民代表大会常务委员会第十七次会议第二次修订）

Revising the Law on the Prevention and Control of Solid Waste Pollution (2020.9.1)

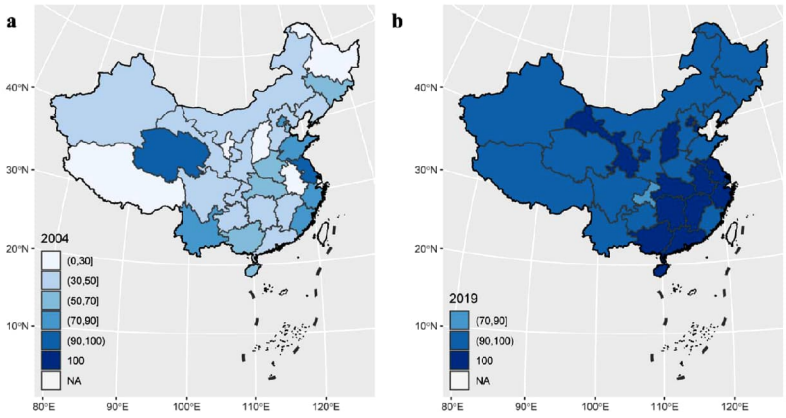
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Evolution of Domestic Waste Treatment in China

Evolution of Domestic Waste Treatment Methods



Harmless Treatment Rate of Domestic Waste (%) of 31 regions in China

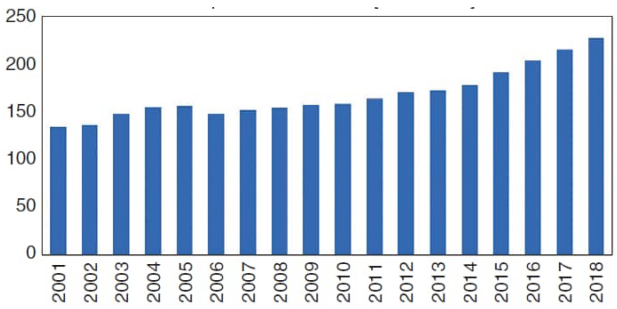


(Guo et al, 2021)

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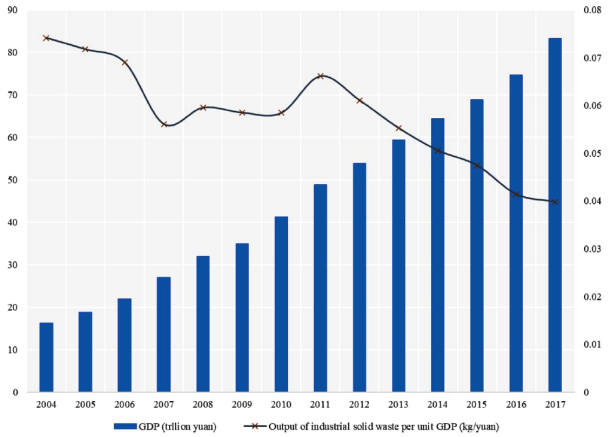
Evolution of Domestic Waste Treatment in China

Production of Municipal Solid Waste in China's Cities (excluding counties, towns and villages)



(Lee et al, 2021)

China's GDP and General Industrial Solid Waste Output per Unit GDP from 2004 to 2017



(Guo et al, 2021)

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Waste Sorting in Shenzhen
可回收物
Recyclable
适宜回收和资源化利用的生活垃圾，包括玻璃、废弃金属、塑料、纸类、废弃家具、织物、电器电子产品等。
塑料
塑料桶
塑料盒
塑料玩具
塑料托盘
沐浴露瓶
塑料饮料瓶
纸类
报纸/杂志
旧书
牛奶盒
纸箱
广告纸
玻璃
平板玻璃
玻璃牛奶瓶
酒瓶
玻璃调味瓶
玻璃杯
金属
易拉罐
刀具
水龙头
金属瓶盖
钥匙
螺丝
剪刀
金属锅
金属容器
家具
床垫
沙发
衣柜
织物
衣服
背包/书包
床上用品
电器电子产品
电视机
洗衣机
冰箱
平板电脑
手机

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Waste Sorting in Shenzhen
有害垃圾
Hazardous Waste
对人体健康或者自然环境造成直接或者潜在危害且应当专门处理的生活垃圾，包括电池、灯管、家用化学品等。
电池
干电池
纽扣电池
充电式电池
灯管
单端紧凑型节能荧光灯
环形荧光灯
直管
家用化学品
化妆品
油漆桶
药丸/片剂
软膏
水银温度计
其他垃圾
Residual Waste
除可回收物、厨余垃圾、有害垃圾以外的其他生活垃圾。
尘土
破损花盆/破碟/陶瓷制品
毛发
烟头
旧毛巾
已污染的塑料袋(膜)
一次性用品
已污染的纸巾
内衣裤
厨余垃圾
Food Waste
容易腐烂的食物残渣、瓜皮果核等含有机质的生活垃圾，包括家庭厨余垃圾、餐厨垃圾、其他厨余垃圾等。
剩饭剩菜
蛋壳
菜帮菜叶
茶渣
果皮

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Practice of "Zero-Waste City" in Shenzhen

- Work Plan for the Pilot Program of "Zero-Waste City" (2018)
- 11+5 cities/areas were selected for the program (2019):

Shenzhen, Guangdong
Baotou, Inner Mongalia
Tongling, Anhui
Weihai, Shandong
Central districts, Chongqing
Shaoxing, Zhejiang
Saya, Hainan
Xuchang, Henan
Xuzhou, Jiangsu
Panjin, Liaoning
Xining, Qinghai
Xiong'an, Hebei
Economic and Technological Development Zone, Beijing
Sino-Singapore Tianjin Eco-city
Guangze County, Fujian
Ruijin, Jiangxi

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Practice of "Zero-Waste City" in Shenzhen

	2020	2025	2035	2050
Volume added of garbage generated per capital per year	Approach to 0	Negative		International leading level, international model of "zero waste city"
Industrial solid waste production intensity	5% lower than 2018	15% lower than 2020	Significantly reduced	
Domestic waste recycling rate	35%	38%	International leading level	
Rate of comprehensive utilization of industrial and solid waste	90%	95%	98%	
Rate of utilization of house demolition waste	90%	95%	98%	

(Implementation Plan of "Zero-Waste" City of Shenzhen，2019)

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Practice of "Zero-Waste City" in Shenzhen

- Waste Generation Reduction
  - Quota system of construction solid waste (建筑垃圾限量排放)

  Standard for Construction & Demolition Waste Discharge Quota of Construction Engineering (建设工程建筑废弃物排放限额标准)

- Improving recycling and utilization rate
  - Waste sorting system
  - Investing in solid waste utilization and treatment facilities and enterprises

- Institutional Support
  - Environmental Pollution Liability Insurance
  - Environmental credit system
  - Smart and grid monitoring and enforcement system

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Practice of "Zero-Waste City" in Shenzhen



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Practice of “Zero-Waste City” in Shenzhen



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Reflective Question:

How does solid waste pollution different from air and water pollution?

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Transferable Negative Externality and Environmental Justice

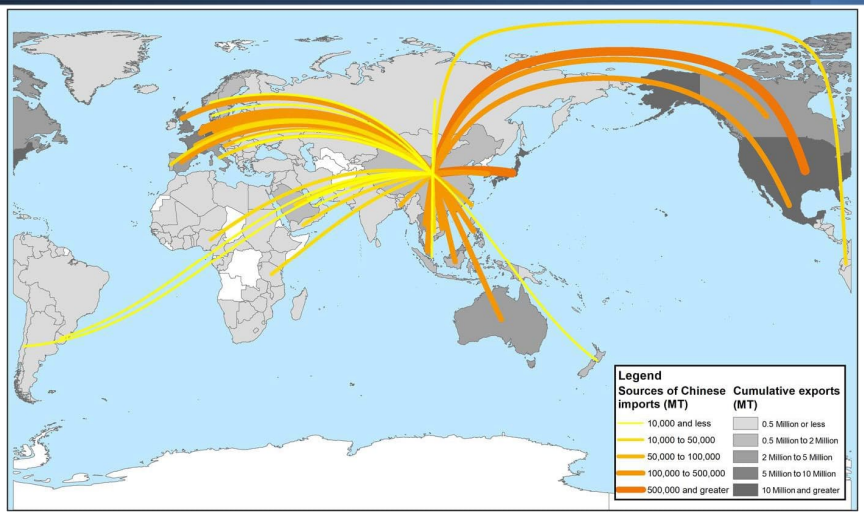
- Environmental Justice: The fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income with respect to the development, implementation and enforcement of environmental laws, regulations and policies.
- Fair Treatment: no group of people should bear a disproportionate share of the negative environmental consequences resulting from industrial, governmental and commercial operations or policies.
- Meaningful involvement: People have an opportunity to participate in decisions about activities that may affect their environment and/or health; The public's contribution can influence the regulatory agency's decision; Community concerns will be considered in the decision-making process; and Decision makers will seek out and facilitate the involvement of those potentially affected.

(US EPA Official Website)



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"Foreign Garbage"



Sources of plastic waste imports into China in 2016 and cumulative plastic waste export tonnage (in million MT) in 1988-2016

(Brooks, Wang, and Jambeck, 2018)

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Ban on Importing Wastes

- 2017.7.18- Releasing "Implementation Plan for Prohibiting the Entry of Foreign Garbage and Advancing the Reform of the Solid Waste Import Administration System" [禁止洋垃圾入境 推进固体废物进口管理制度改革实施方案]
- 2018.1.1- Stopping taking in 24 types of waste[《进口废物管理名录》（2017）]
- 2018.12.31- including 16 more types of wastes into the banning list
- 2019.7.1- Adding more types of waste in the restricting list
- 2020.11.24- Releasing "Announcement regarding Matters Concerning Comprehensively Prohibiting the Import of Solid Waste" [关于全面禁止进口固体废物有关事项的公告]
- 2021.1.1- prohibiting the entry of all forms of solid wastes

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Ban on Importing Wastes



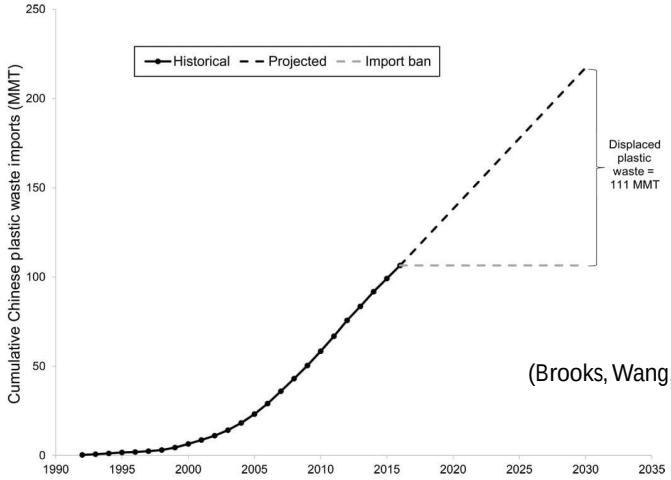
China's ban on foreign solid waste in effect



China's Trash ban Creates Crisis for US Recyclers

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Impacts of the Ban

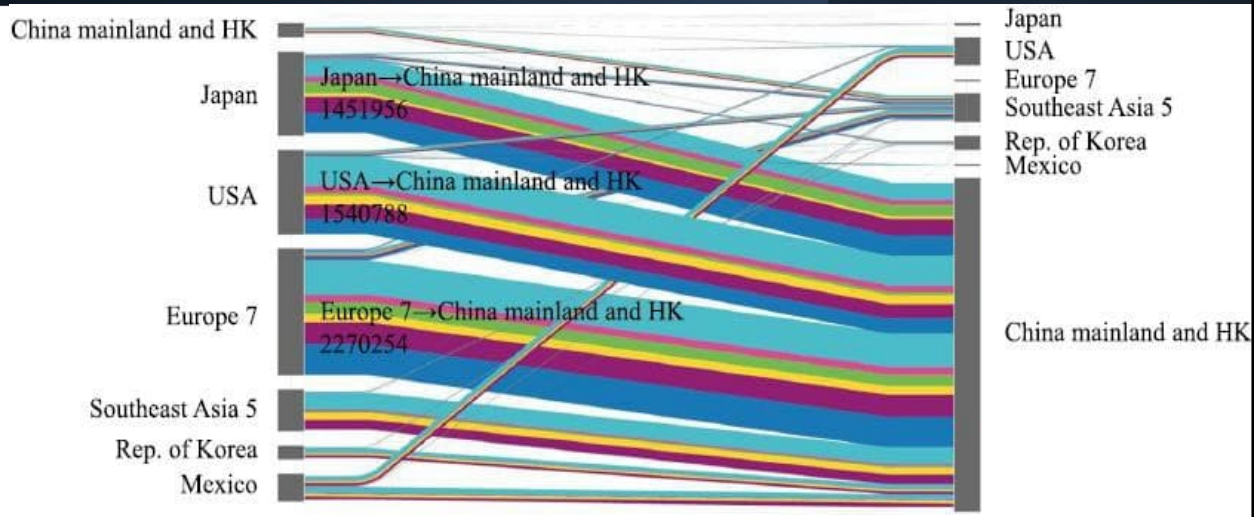


(Brooks, Wang, and Jambeck, 2018)

Estimated mass of global displaced plastic waste due to the new Chinese import ban on cumulative imports of PE, PS, PVC, and other plastics into China

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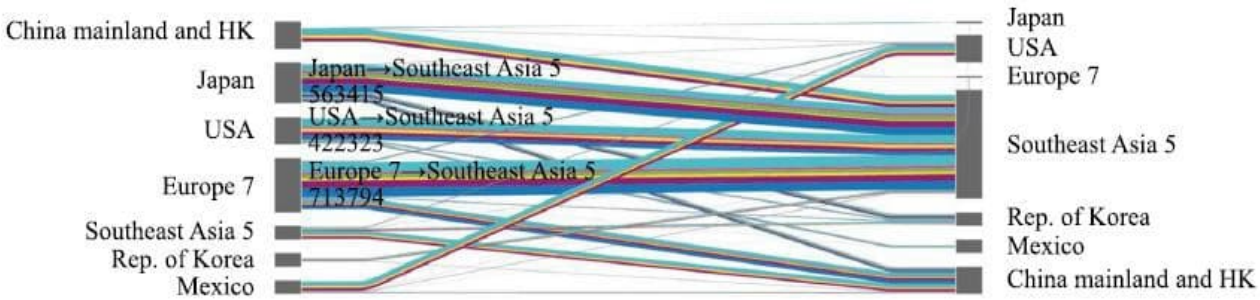
Impacts of the Ban



The Global Trade Flows of Six Types of Plastic Waste Prior to the Ban (Wen, Xie, Chen, and Dinga, 2021)

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Impacts of the Ban



The Global Trade Flows of Six Types of Plastic Waste after the Ban (Wen, Xie, Chen, and Dinga, 2021)

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Discussion:

We have learned the practices on solid waste management in Sweden, China, and the US. What is your opinion on plastic ban and imported solid waste?

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